# “SMARTCINNAMON”: A MULTI-MODAL MACHINE LEARNING AND IOT FRAMEWORK FOR PRECISION CINNAMON CULTIVATION

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## DECLARATION

We declare that this is our work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date

Signature of the Co - supervisor: Date

## ABSTRACT

*Cinnamon Plantation Vacant Area Detection and Yield Forecasting*

Ceylon cinnamon is one of Sri Lanka's most valuable export crops, but plantation productivity is often limited by underutilized land, manual surveying errors, and lack of accurate yield prediction methods. Traditional land management techniques rely on visual inspections and GPSbased tools, which are prone to inaccuracies caused by poor signal quality, weather interference, and human error. This study proposes an AI and GIS-based solution for cinnamon plantation management, focusing on vacant area detection, tree counting, and yield forecasting. The system integrates satellite/drone imagery with a mobile/web platform, where images are analyzed using three machine learning models: a Convolutional Neural Network (CNN) for classification, a YOLO model for tree detection, and a U-Net segmentation model for vacant-area identification. GIS tools further enhance precision by calculating plantation boundaries and available plantable space.

Processed data is presented on an interactive dashboard that provides tree counts, vacant land distribution, area measurements, and projected bark yield (kg). Farmers and export companies can use these insights to plan plantation expansion, optimize spacing, and align harvest forecasts with market demands. Anticipated results include improved land utilization, reduced resource wastage, and more reliable yield predictions. The system is expected to strengthen the cinnamon value chain by enabling data-driven decisions, supporting sustainable cultivation, and ensuring greater profitability for both small-scale farmers and large exporters.

**Keywords: Cinnamon plantation, vacant area detection, yield forecasting, machine learning, GIS analysis.**

## LIST OF ABREVATIONS

| | |
|---|---|
| GIS | Geographic Information System |
| FLUTTER | Cross-platform UI toolkit for mobile/web/ desktop |
| CNN | Convolutional Neural Network |
| YOLO | You Only Look Once (Object Detection Model) |
| U-Net | Convolutional Neural Network for Image Segmentation |
| ML | Machine Learning |
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| DB | Database |

| | |
|---|---|
| WBS | Work Breakdown Structure |
| SDLC | Software Development Life Cycle |
| AWS | Amazon Web Services |

*Table I: List of Abbreviators*

## *1. INTRODUCTION*

Ceylon cinnamon is a major export for Sri Lanka. Estate managers must measure fields, see crop status, and plan replanting clearly. Old methods—field walks and hand-drawn maps—are slow, costly, and can miss small empty spaces. With remote sensing and GIS, estates can turn new drone or satellite images into accurate, repeatable maps for decision-making [1], which improves both speed and reliability.

General mapping tools can still struggle in real plantations. Image quality, clouds, shadows, and sensor differences can reduce classification accuracy [2] and lead to poor land allocation or wasted inputs. A structured RS-GIS workflow that uses tested land-use/land-cover methods helps cut these errors and produces auditable results for managers [2], [3].

Planning the economics matters as much as mapping. Managers need to know how many trees they can add, where to place them, and what yield to expect. GIS brings spatial data together with decision layers so teams can plan land use and budgets more effectively [4], supporting better replanting and expected returns.

To solve these needs, we propose the **Cinnamon GIS Yield Planner**—a simple, map-based workflow tailored to cinnamon. The system (i) acquires drone or high-resolution satellite imagery, (ii) applies image segmentation and ML to classify "cultivated" vs. "vacant" areas, (iii) simulates replanting layouts at a chosen spacing (e.g., 2 m × 2 m) to compute plantable points, and (iv) estimates added yield from tree counts and historic bark productivity. The classification follows well-known LULC practices (e.g., supervised/segmentation approaches) to keep results consistent and transparent [2], [3], while the pipeline stays cinnamon-specific and estate-ready [1].

Final results appear on an interactive dashboard so managers can read

"**X** new trees can be added → **Estimated Yield +Y kg**."

This turns imagery into clear replanting actions [1], reduces the need for manual surveying [2], and builds confidence in planning decisions.

## 1.1. BACKGROUND AND LITERATURE SURVEY

Accurate field measurement and land management are essential for commercial plantations. Manual field walks and paper maps are slow, labor-intensive, and can miss small gaps inside blocks. Remote sensing (RS) and Geographic Information Systems (GIS) now let estates turn recent aerial or satellite images into repeatable maps for planning, reducing human error and cost [2]. Integrating RS layers with plantation GIS (roads, blocks, streams, elevation) gives managers a single view for decisions instead of relying on one image or ad-hoc sketches [2].

A consistent finding across the literature is that **integration + validation** matter as much as imagery quality. RS outputs should be brought into GIS and then checked with clear accuracy metrics—Overall Accuracy, User's/Producer's Accuracy, and Kappa—before operational use [2]. For land-cover mapping, **supervised classification** (commonly Maximum Likelihood) remains a practical baseline, with performance reported via confusion matrices so end-users understand omissions/commissions [6]. These practices make maps auditable for estate managers and external stakeholders [2], [6].

Vegetation indices help separate canopy from bare ground. Indices such as **NDVI** and **MSAVI2** highlight photosynthetic activity and reduce soil background effects, improving the distinction between productive cinnamon canopy and low-vegetation/vacant patches [5], [6]. In widely used agricultural schemes, **perennial plants** explicitly include cinnamon alongside tea, coffee, rubber, and banana, confirming that cinnamon canopy is a legitimate, mappable class in standard LULC taxonomies [5]. For a cinnamon estate, this means vacancy detection can be framed as "perennial canopy vs. non-canopy," supported by well-known spectral cues [5], [6].

In Sri Lanka, prior studies have already mapped **cinnamon cultivation change** using aerial photographs, satellite images, and GPS ground points, underscoring cinnamon's export value and the role of spatial evidence for decision-making [1]. Those works establish feasibility and data availability but stop short of plantation-ready analytics—specifically, they do not detect **withinblock vacant** areas with modern segmentation/ML, generate **spacing-aware replanting** layouts, or (iii) translate added trees into **yield forecasts** for managers [1]. This unaddressed need motivates a cinnamon-specific pipeline.

From a planning perspective, GIS is more than a mapping tool; it is a **decision platform**. Standard land-use planning practice brings multiple criteria (access, slope, buffers, water, soil) into one analysis so managers can prioritize interventions spatially [4]. After classification,

planners can rank candidate areas for replanting using simple rules or formal MCDA approaches inside GIS, aligning with how agricultural protection and allocation studies proceed in other regions [5]. This mirrors estate realities where spacing, access, erosion risk, and logistics all matter alongside pure area figures [4], [5].

Change detection is also central to estate monitoring. Reviews recommend **post-classification comparison**—classifying each date separately and then comparing the maps—because it reduces sensitivity to atmospheric/sensor differences between acquisition dates and produces intelligible "from–to" change matrices for management reports [2]. For cinnamon, this enables periodic audits of cultivated vs. vacant area and quantifies the impact of replanting programs season-toseason [2], [6].

Final spatial outputs should link to **production economics**. Studies show how RS-derived landcover and vegetation metrics are used to support crop planning and resource allocation, providing a bridge from "map pixels" to **yield-oriented decisions** [3]. For cinnamon estates, combining vacancy-based **tree counts** (driven by block-level spacing) with **historic bark productivity** and **survival assumptions** turns a classification map into business metrics—"X trees can be added → +Y kg expected yield"—that are usable by estate managers and exporters [3], [4], [5].

**Implication for this study.** The literature supports a practical workflow: (1) acquire drone/ satellite imagery; (2) classify cinnamon canopy vs. vacant using supervised/segmentation methods with NDVI/MSAVI2 aids [5], [6]; (3) validate accuracy with confusion-matrix metrics [2], [6]; (4) run GIS-based spacing and suitability to generate replanting points [4], [5]; and (5) convert added trees to yield/revenue using estate parameters [3]. This closes the gap between regional cinnamon mapping [1] and an operational, plantation-level **Cinnamon GIS Yield Planner**.

## *2. RESEARCH GAPS*

GIS and remote sensing have become very useful tools for agriculture, especially for mapping land use and detecting changes over time. Studies have used satellite data to monitor forests, water bodies, urban expansion, and crop areas with good accuracy. In Sri Lanka, research has also been carried out on cinnamon lands to identify how the total cultivated extent has reduced due to urbanization and land fragmentation.

These findings show the value of GIS in agricultural planning. However, most of these studies stop at general land change analysis and do not focus on identifying *vacant areas inside existing plantations*. For cinnamon, vacant area planting is a major part of long-term productivity, but no published studies have tried to use GIS and machine learning to calculate available planting space within plantations. Another limitation is that current GIS-based agricultural studies do not provide automated replanting layouts using plantation spacing standards such as 2 m × 2 m, which is important for planning large-scale cinnamon estates. In addition, while yield forecasting has been explored for other crops using remote sensing and GIS, there is no system that links vacant area detection with cinnamon-specific yield prediction based on tree counts and bark productivity data.

Finally, most existing approaches still depend heavily on manual surveys and ground measurements, which are slow, costly, and sometimes inaccurate. Because of these gaps, there is a clear need for a new solution that integrates drone or satellite imagery, machine learning segmentation, and GIS-based planning to automatically detect vacant areas in cinnamon plantations, generate optimized replanting layouts, and forecast yield in a single system.

*Table II : Research Gap*

| **Conducted By** | Accurate Measurement | Vacant Area DetecXon | Yield ForecasXng | InteracXve GIS Tools | Customizable ReplanXng Layout |
|---|---|---|---|---|---|
| Research A [1] | ✓ | | | | |
| Research B [3] | ✓ | | | | |
| Research C [4] | ✓ | ✓ | | | |
| Research D [5] | ✓ | ✓ | | | |
| Research E [6] | ✓ | | ✓ | | |
| Proposed System | ✓ | ✓ | ✓ | ✓ | |

### *3. RESEARCH PROBLEM*

Cinnamon plantation managers and estate supervisors face major challenges in efficiently identifying and utilizing vacant areas within their lands. Current GIS and remote sensing applications mainly provide information on total land cover or general crop extent, but they do not offer tools to detect unused or vacant spaces inside existing plantations. As a result, replanting decisions are often based on manual surveys or rough visual checks, which are timeconsuming, labor-intensive, and prone to human error. Furthermore, yield forecasting in cinnamon plantations is mostly based on rough historical averages instead of accurate plant counts or spatial analysis, which creates uncertainty in production planning and reduces the reliability of export forecasting.

**Do you face problems with detec5ng vacant planta5on areas and forecas5ng accurate cinnamon yields?**

According to a recent survey conducted among plantation supervisors and estate managers, 79.41% reported difficulties in identifying exact vacant areas within their cinnamon plantations, while 82.35% stated that yield projections are often unreliable due to lack of accurate plant counts. These results show a clear limitation in current tools, which mostly provide only basic land measurement or general crop monitoring. This emphasizes the urgent need for a system that combines drone/satellite imagery, machine learning–based classification, and GIS analysis to accurately map vacant areas, plan replanting layouts, and forecast yields with higher reliability.

According to a recent survey conducted among plantation supervisors and estate managers, 79.41% reported difficulties in identifying exact vacant areas within their cinnamon plantations, while 82.35% stated that yield projections were often unreliable due to lack of accurate plant counts. These results highlight a serious gap in the current systems, which mainly provide only basic mapping functions. This clearly emphasizes the urgent need for a solution that integrates drone/satellite imagery, machine learning–based classification, and GIS analysis to accurately detect vacant areas, simulate replanting layouts, and forecast yields with higher reliability.

In addiXon to vacant-area mapping, cinnamon growers also struggle with replanXng layout planning. TradiXonal methods rely on approximate spacing, such as 2 m × 2 m, but this is done manually without considering exact land availability or plantaXon geometry. This ohen results in irregular spacing, inefficient land usage, and reduced producXvity.

Do you face difficulXes in planning replanXng layouts for cinnamon plantaXons?

n a survey, 76.47% of growers reported problems in designing uniform replanting layouts, often leading to overcrowding or under-utilized land. This indicates the need for an automated tool that can calculate optimal replanting patterns and visually guide managers on how many plants can be added in each vacant section.

Another significant challenge faced by cinnamon plantation managers is the lack of integrated yield and revenue forecasting. Current tools do not combine spatial data (such as plant counts and vacant area recovery) with historical productivity trends, making it difficult to plan exports, manage resources, or predict long-term profitability. Without reliable yield forecasting, estate managers cannot make accurate financial and operational decisions, leading to inefficiencies and revenue loss.

Are you facing challenges with predicXng cinnamon yields and long-term revenue potenXal?

In the same survey, 88.23% of estate managers reported that their yield predictions were frequently inaccurate or delayed, creating difficulties in both production and export planning. This highlights the need for a GIS-based solution that integrates land vacancy detection, automated replanting layouts, and machine learning–driven yield forecasting to support precise decision-making.